MIL-R-50858C(MI)
7 November 1979
Superseding
MIL-R-50858B(MI)
31 May 1974

MILITARY SPECIFICATION

ROCKET, PRACTICE, 35MM, SUBCALIBER, M73,

PARTS FOR LOADING AND ASSEMBLING

This specification is approved for use by the US Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification contains requirements not normally required by the drawings, and provides the quality assurance provisions for fabrication of parts, loading, assembling and packing for one type of complete practice rocket designated as Rocket, Practice, 35MM Subcaliber, M73.

2. APPLICABLE DOCUMENTS

* 2.1 Issue of documents. The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)

Military

MIL-P-223	Powder, Black
MIL-P-2496	Primer, Percussion, M29A1, Parts and Loading, Assembling and Packing
MIL-A-2550	Ammunition and Special Weapons; General Specification for
MIL-C-45662	Calibration System Requirements

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Missile Command, ATTN: DRSMI-ED, Redstone Arsenal, AL 35809 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1340

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MIL-P-50981	Propellant Charge Assembly for Rocket, Practice, 35MM, Subcaliber, M73
MIL-P-60048	Primer, Stab, M26; Parts for
MIL-P-60049	Primer, Stab, M26; Loading Assembling and Packing

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-171	Finishing of Metal and Wood Surfaces
MIL-STD-1168	Lot Numbering of Ammunition
MIL-STD-1235	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

US Army

10242725	Rocket, Practice, 35MM Subcaliber, M73
10242743	Carton, Packing, Ammunition, for Rocket, Practice, 35MM, Subcaliber, M73
10242697	Box, Packing, Ammunition, for Rocket, Practice, 35MM, Subcaliber, M73
P-2905	Impact Testing Machine for Explosives

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PUBLICATIONS

ET-10242725 Equipment Tabulation

* (Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids, or request for proposal, shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-B117 - Salt Spray (Fog) Testing, Standard Method of

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

Technical Society and Technical Association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.

3. REQUIREMENTS

3.1 Material. Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Assembly. The M73 practice rocket shall comply with all of the requirements specified on drawing (dwg.) 10242725, and with all of the requirements specified in all of the applicable specifications.

3.3 Black powder. The black powder used in the manufacture of the igniter and ignition transmission line shall conform to all of the requirements of the applicable drawings and MIL-P-223. The moisture content of the black powder in the complete igniter and primer molded assembly shall not exceed 0.50 percent (%).

3.4 Primer, stab, M26. The M26 stab primer used in the fabrication of this rocket shall conform to all of the requirements of MIL-P-60048 and MIL-P-60049. These primers shall not be assembled into the primer holders if two or more years have elapsed since original acceptance of the primers, or if two or more years have elapsed since the last successful check test for deterioration, or if the packaging or packing show evidence of having been subjected to adverse environmental conditions, until the required check test is performed as specified in MIL-P-60049. (See 6.6).

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3.5 Primer, percussion, M29A1. The M29A1 percussion primers used in the fabrication of this rocket shall conform to all of the requirements of MIL-P-2496. These primers shall not be assembled into the molded igniter and primer assembly if two or more years have elapsed since original acceptance of the primers, or if two or more years have elapsed since the last successful check test for deterioration, or if the packaging or packing show evidence of having been subjected to adverse environmental conditions, until the required check test is performed as specified in MIL-P-2496. (See 6.6).

3.6 Rocket propellant charge assemblies. The rocket propellant charge assemblies used in the fabrication of this rocket shall conform to all of the requirements of MIL-P-50981. After removal of the rubber band and assembly to the suspension plates, the propellant grains shall not be broken, cracked or chipped.

3.7 Hydrogen embrittlement of safety clip and fuze spring. The safety clip and fuze spring shall be free of hydrogen embrittlement as evidenced by not breaking when extended to the point of minimum permanent deformation.

3.8 Identification of molded parts. The molds used in the production of molded parts shall leave an identifying mark on the surface of the molded part to identify the mold cavity from which it came. This identification shall be located so as not to become obliterated, removed in subsequent operations and, when possible, be visible after assembly.

3.9 Protective finish. Components requiring protective coatings in accordance with MIL-STD-171 and QQ-P-416 shall show no evidence of white corrosion products of cadmium after 96 hours of exposure to a 5% solution of salt spray.

3.10 Closure.

3.10.1 Yield strength. The closure shall not be permanently deformed beyond the applicable drawing dimensions after having been subjected to an internal hydrostatic pressure of not less than 11,500 nor more than 12,000 pounds per square inch (psi) applied to the threaded cavity and applied in such a manner so as not to restrict deformation.

3.10.2 Ultimate strength. The closure shall not rupture and the threads shall not fail while being subjected to an internal hydrostatic pressure of 15,000 psi minimum applied to the threaded cavity.

3.11 Motor Case.

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3.11.1 Yield strength. The motor case shall not be permanently deformed beyond the applicable drawing dimensions after having been subjected to an internal hydrostatic pressure of not less than 14,000 nor more than 16,000 psi applied to the propellant chamber from the threaded end to as close to the nozzle throat as possible and applied in such a manner so as not to restrict deformation.

3.11.2 Ultimate strength. The motor case shall not rupture and the threads shall not fail while being subjected to an internal hydrostatic pressure of 18,000 psi minimum, applied to the propellant chamber from the threaded end to as close to the nozzle throat as possible and applied in such a manner so as not to restrict deformation.

3.12 Resistance of igniter flash tube to stress cracking. The flash tube used in the fabrication of the molded igniters shall contain no cracks after having been stressed and submerged in "Stress Crack Reagent IGEPAL CO-630" or substantial equal for a minimum of 24 hours. Stressing of the flash tube is accomplished by heating the tube for 10 minutes at a temperature of 160 degrees Fahrenheit (°F) and immediately bending the tube 180 degrees until the ends are parallel and approximately one-half (1/2) inch apart and held in this configuration for the entire time of submersion in the stress crack reagent.

3.13 Igniter.

3.13.1 Strength of assembly. The flash tube of the igniter and primer molded assembly shall not become disassembled from either end while the assembly is being subjected to a static force of 40 pounds minimum, applied along the longitudinal axis of the flash tube in a direction to separate the components applied to the molded cup at one end and the primer block at the other. The heat seal on the lip of the igniter cup shall not leak when subjected to 10 psi or air pressure.

* 3.13.2 Static functioning. The igniter, when seated in the nozzle and sealed as specified on drawing 10242725, and with an insert representing the volume of the complete propellant charge and suspension plate placed in the motor case shall, when functioned by impacting the primer, produce a pressure/time trace which will measure peak pressure and rise time. Peak pressure is defined as the highest point on the pressure/time trace. Rise time is defined as the elapsed time interval from when the primer is initiated to 90% of peak pressure. At all temperatures from minus (-) 10° to plus (+) 135°F, the peak pressure shall not be less than 300 psi and shall not exceed 900 psi, and the rise time shall not be less than 1.5 milliseconds (ms) and shall not exceed 50 ms. The molded cup of the igniter shall not rupture or become disassembled from the motor case. Venting by failure of the flash tube is normal for this mode of functioning. The polyethylene igniter, prior to seating in the nozzle, may be electrostatically treated for adhesiveness.

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3.13.3 Waterproofness. The igniter portion of the assembly shall function satisfactorily after the complete assembly has been submerged in water at a minimum depth of 3 inches for a minimum of 24 hours while the water temperature is maintained between 60° and 90°F. Satisfactory functioning is defined as functioning between 60°F and 90°F on the first attempt with a rise time not in excess of 50 ms.

3.14 Sensitivity of blended flash mix. The flash mix blended in the head assemblies shall not ignite nor detonate when a specimen is impacted by the 2 kilogram weight of the Picatinny Arsenal impact test apparatus (See dwg P-2905) dropped from a height of 8 inches minimum.

3.15 Closure assembly. Radiographic examination of the assembly shall verify: that the safety clip is complete and protrudes adequately into the closure assembly; that no components are missing; and that the inertia weight and firing pin are not latched. The inertia weight and the firing pin are latched when the spring is compressed to nearly solid height and the point of the firing pin protrudes through the front face of the inertia weight.

3.16 Complete practice rocket.

3.16.1 Complete round functioning. The complete rocket shall function satisfactorily at any temperature from -10°F to +135°F. The rocket shall not fail to function on the first attempt, shall not experience ignition delay in excess of 1 second from activation of the trigger mechanism to the emergence of the rocket from the muzzle of the launcher and shall completely consume the propellant within the launcher. Heads shall function on impact with 3/16 inch thick steel plate at angles of 90° to 26° to the line of sight and the event shall be visible under normal daylight conditions from 300 meters.

3.16.2 Rocket Integrity. The rocket shall not suffer motor/closure failure or rupture or premature functioning of the head flash mix prior to graze or target impact.

3.16.3 Rocket accuracy at 70°F. When conditioned at 70°F and fired from a fixed mount, the rocket shall impact a 10 ft. by 10 ft. vertical target placed 200 meters from the launcher muzzle such that the average center of impacts plus or minus the product of 1.5 and the sample standard deviation from center target does not exceed 60 inches.

3.16.4 Waterproofness. The rocket shall function satisfactorily (accuracy excepted) when fired at 70°F after being submerged in water to a depth of three feet for two hours, plus or minus 10 minutes.

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3.16.5 Rocket handling safety. The rocket shall not become propulsive and shall be safe to fire when dropped bare one (1) time each, first on the tail, then on the head, from a height of five (5) feet onto concrete with the safety clip removed. The head shall not function. Parts of the rocket shall remain secure, although breakage of exposed plastic components is permissible, and the rocket shall not allow explosive components to be exposed.

3.17 First article inspection. This specification makes provisions for first article inspection. Requirements for submission of first article samples shall be as specified in the contract.

3.18 Workmanship. All parts, subassemblies and assemblies shall be free of chips, dirt, grease, rust and any other foreign material that may impair proper functioning of the item, or that may cause premature functioning or deterioration in storage. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent used. Painted or coated surfaces shall be restored when these surfaces have been marred or scratched. All markings shall be clear and sharply defined. Parts shall be free of cracks, breaks, chips or other visible defects.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product. At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

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a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with an identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The intitial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection.

4.2.1 Submission. Prior to initiating regular production, the contractor shall submit a first article sample as designated by the Contracting Officer for evaluations in accordance with the provisions of 4.2.2. The first article sample shall consist of 20 of each assembly, sub-assembly and component comprising the M73 rocket and sufficient test samples to fulfill the test requirements of 4.3.3. The first article sample shall consist of material that has been produced by the contractor or furnished by a supplier and manufactured using the same production processes, procedures and equipment that will be used in fulfilling the contract. The first article sample, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process material used, drawing, specification or source of supply so as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects shall not be submitted unless specifically authorized by the Contracting Officer.

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4.2.2 Inspection to be performed. Assemblies or components will be subjected by the Government to any or all of the examinations or tests specified in 4.3.2 and 4.3.3 of this specification and any or all of the requirements of the applicable drawing.

4.2.3 Rejection. The first article sample shall be rejected if, in the igniter static functioning test (see 4.3.3.1.13) one or more duds are observed, or if one or more igniter cups become disassembled, or if one or more igniters exhibit nonconforming rise times, or if two or more igniters exhibit nonconforming peak pressure. Also, the first article sample shall be rejected if six (6) or more rockets miss the target in the combined sample of fifty-five (55) complete rounds tested as required in 4.3.3.1.15.1 and 4.3.3.1.15.4. The first article sample shall be rejected if any assembly, sub-assembly or any component fails to comply with any of the applicable requirements. The Government reserves the right to terminate its inspection upon observing any nonconformance of the sample to the specified requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article sample or portion thereof. Until a first article sample is accepted, the contractor is in no way authorized by the Government to initiate regular production unless otherwise directed by the Contracting Officer.

4.3 Quality conformance inspection.

4.3.1 Lot formation. The term "inspection lot", as used in this specification, is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions, and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specifications or drawings, not affecting safety, performance, interchangeability or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each lot shall contain:

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a. Propellant assemblies from not more than one lot from one manufacturer.

b. Parts and parts assemblies from not more than one interfix lot number from one manufacturer. (Applicable to each part or parts assembly individually).

c. Igniter and primer molded assemblies from not more than one interfix lot number from one manufacturer. Each igniter and primer molded assembly lot shall contain primers from not more than two lots from one manufacturer and black powder from not more than two lots from one manufacturer.

d. Fin and motor case assemblies from not more than two lots from one manufacturer.

4.3.2 Examination. The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for Critical Defects (and Major Defects, when so specified) shall be 100 percent. Sampling plans for Major and Minor Defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of Major and Minor Defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each Major Defect and and AQL of 0.65 percent for each Minor Defect. When this method is elected, it shall be applied to all of the Major and Minor Defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Head, molded (see 3.8 and dwg. 10242704, covering a detail of dwg. 10242708).

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Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical: None defined.			
Major:	AQL 0.65 Percent		
101.	Concentricity of lugs with outside diameter of head	Gage	01001
102.	Diameter over lugs	Gage	01002
103.	Length to bottom of large inside diameter, maximum (max.)	Gage	01003
104.	Component incomplete or mold identification missing	Visual	01004
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.18)	Visual	01005

4.3.2.2 Head assembly, loading (see dwg. 10242708, covering a detail of dwg. 10242690).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Loose composition mix on exposed surfaces	Visual	02001
2.	Disc not properly sealed or damaged to the extent that the explosive charge may leak out	Visual	02002
Major:	AQL 0.40 Percent		
101.	Composition mix powder missing or incomplete	Visual	02003
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.18)	Visual	02004

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4.3.2.3 Closure, after 100% hydrostatic testing (See dwg. 10242705, covering a detail of dwg. 10242710).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 Percent		
101.	Min. length of effective threads	Gage	03001
102.	Minor diameter, max	Gage	03002
103.	Pitch diameter, max	Gage	03003
104.	Depth of primer holder cavity, minimum (min)	Gage	03004
105.	Concentricity of fuze cavity with primer holder cavity	Gage	03005
106.	Location of safety clip hole	Gage	03006
107.	Concentricity of threads with outside diameter	Gage	03007
108.	Concentricity of small outside diameter to large outside diameter	Gage	03008
Minor:	AQL 4.50 Percent		
201.	Length of small outside diameter, min.	Gage	03009
202.	Diameter of fuze cavity, min.	Gage	03010
203.	Protective finish missing or inadequate	Visual	03011
204.	Evidence of poor workmanship (See 3.18)	Visual	03012

4.3.2.4 Clip, Safety (see dwg. 10242683, covering a detail of dwg. 10242710).

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Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 1.00 Percent		
101.	Wire diameter, Min.	Gage	04001
102.	Length of arm, min.	Gage	04002
103.	Inside diameter, max.	Gage	04003
104.	Location of arm	Gage	04004
105.	Protective finish incomplete or inadequate	Visual	04005
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.18)	Visual	04006

4.3.2.5 Weight, inertia (see dwg. 10242747, covering a detail of dwg. 10242710).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 1.00 Percent		
101.	Diameter of hole	Gage	05001
102.	Width of slot. min.	Gage	05002
103.	Dimension from small outside diameter to bottom of slot, min.	Gage	05003
104.	Chamber missing from semicircular hole	Gage	05004
105.	Protective finish incomplete or inadequate	Visual	05005
Minor:	AQL 1.00 Percent		
201.	Diameter of large outside diameter, max.	Gage	05006
202.	Evidence of poor workmanship (see 3.18)	Visual	05007

4.3.2.6 Holder, primer (see dwg. 10242699, covering a detail of dwg. 10242702.

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Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 Percent		
101.	Outside diameter	Gage	06001
102.	Large inside diameter, max.	Gage	06002
103.	Concentricity of large inside diameter with outside diameter	Gage	06003
104.	Depth of large inside diameter	Gage	06004
105.	Flash hole missing	Visual	06005
106.	Protective finish incomplete or inadequate	Visual	06006
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.18)	Visual	06007

4.3.2.7 Spring, weight (see dwg. 10242746, covering a detail of dwg. 10242710).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 Percent		
101.	Improper location of tang	Gage	07001
102.	Spring force not in working range	Gage	07002
103.	Solid height, max.	Gage	07003
104.	Spring fails to slide freely in test cavity in max. compressed condition	Gage	07004
105.	Protective finish incomplete or inadequate	Visual	07005
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.18)	Visual	07006

4.3.2.8 Pin, firing (see dwg. 10242745, covering a detail of dwg. 10242710).

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Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 Percent		
101.	Profile of point	Gage	08001
102.	Concentricity of outside diameter with diameter of shank	Gage	08002
103.	Length from large diameter to smallest diameter, min.	Gage	08003
104.	Length of smallest diameter, min.	Gage	08004
105.	Length from smallest diameter to pin point, min.	Gage	08005
106.	Diameter of smallest diameter, max.	Gage	08006
107.	Protective finish incomplete or inadequate	Visual	08007
Minor:	AQL 1.00 Percent		
201.	Large diameter, max.	Gage	08008
202.	Evidence of poor workmanship (See 3.18)	Visual	08009

4.3.2.9 Assembly, primer holder (see dwg. 10242702, covering a detail of dwg. 10242710).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 Percent		
101.	Primer inverted, improperly staked or deformed	Visual	09001
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.18)	Visual	09002

4.3.2.10 Assembly, closure (see dwg. 10242710, covering a detail of dwg. 10242690).

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Categories	Defects	Method of Inspection	Code No.
Critical:	None defined:		
Major:	AQL 0.40 Percent		
101.	Primer holder assembly inadequately staked or inverted	Visual	10001
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.18)	Visual	10002

4.3.2.11 Head and closure assembly (see dwg. 10242690, covering a detail of dwg. 10242725).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 Percent		
101.	Assembly loose	Manual	11001
102.	Safety clip not sealed in place	Visual	11002
103.	Foreign matter inside threaded closure cavity	Visual	11003
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.18)	Visual	11004

4.3.2.12 Plate, suspension (see dwg. 10242684, covering a detail of dwg. 10242725).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 Percent		
101.	Diameter of holes	Gage	12001
102.	Outside diameter of plate, max.	Gage	12002
103.	Location of holes	Gage	12003
104.	Depth of Counterbore	Gage	12004
105.	One or more slots missing	Visual	12005
106.	Protective coating missing or incomplete	Visual	12006
Minor:	AQL 1.00 Percent		
201.	Thickness, min.	Gage	12007
202.	Evidence of poor workmanship (see 3.18)	Visual	12008

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4.3.2.13 Motor Case after 100% hydrostatic testing (see dwg. 10242706, covering a detail of dwg. 10242720).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 Percent		
101.	Pitch diameter, min.	Gage	13001
102.	Length of large inside diameter to to radius at nozzle approach angle, min.	Gage	13002
103.	Throat diameter	Gage	13003
104.	Concentricity of thread with outside diameter	Gage	13004
Minor:	AQL 1.50 Percent		
201.	Diameter at end of nozzle	Gage	13005
202.	Major diameter of thread, min.	Gage	13006
203.	Protective coating incomplete or inadequate	Visual	13007
204.	Evidence of poor workmanship (see 3.18)	Visual	13008

4.3.2.14 Fin and Motor case assembly (see 3.8 and dwg. 10242720, covering a detail of dwg. 10242696).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 Percent		
101.	Diameter over fins	Gage	14001
102.	Concentricity of diameter over fins with outside diameter of motor case	Gage	14002
103.	Fins incomplete or have burrs or flashing	Visual	14003
104.	Mold identification missing	Visual	14004
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.18)	Visual	14005

4.3.2.15 Housing, primer (see dwg. 10048796, covering a detail of dwg. 10242696).

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Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 Percent		
101.	Concentricity of primer cavity with large outside diameter	Gage	15001
102.	Diameter of primer cavity, max.	Gage	15002
103.	Diameter of hole, min.	Gage	15003
104.	Depth of primer cavity	Gage	15004
Minor:	AQL 1.00 Percent		
201.	Total height, min.	Gage	15005
202.	Evidence of poor workmanship (see 3.18)	Visual	15006

4.3.2.16 Cup, igniter, molded assembly (see 3.8 and dwg. 10242695, covering a detail of dwg. 10242744).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 Percent		
101.	Profile of exterior of cup from large end to raised rib	Gage	16001
102.	Large inside diameter, min.	Gage	16002
103.	Length of large inside diameter, min.	Gage	16003
104.	Length to end of flash tube in bottom of large cavity, min.	Gage	16004
105.	Mold identification missing, component incomplete, tube kinked, cracked or assembly damaged	Visual	16005
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.18)	Visual	16006

4.3.2.17 Line, transmission, ignition (see dwg. 10048801, covering a detail of dwg. 10242744.)

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Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 Percent		
101.	Explosive missing or incomplete	Visual	17001
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.18)	Visual	17002

4.3.2.18 Sub-assembly, igniter, loading, prior to assembly of black powder (see dwg. 10242744, covering a detail of dwg. 10242696).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 Percent		
101.	Length of transmission line in igniter cup, min.	Gage	18001
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.18)	Visual	18002

4.3.2.19 Sub-assembly, igniter, loading; prior to heat seal of lid (see dwg. 10242744, covering a detail of dwg. 10242696).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 Percent		
101.	Weight of black powder	Balance	19001
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.18)	Visual	19003

4.3.2.20 Assembly, igniter and primer molded (see 3.8 and dwg. 10242696, covering a detail of dwg. 10242725).

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Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Primer inverted, damaged or above flush	Gage	20001
2.	Disc or parts loose or damaged to the extent that the explosive charge may leak out	Visual	20002
3.	Loose explosive in motor case	Visual	20003
Major:	AQL 0.65 Percent		
101.	Outside diameter of primer holder section, max.	Gage	20004
102.	Thickness of primer holder section	Gage	20005
103.	Angle between primer block and flash tube	Gage	20006
104.	Mold identification missing	Visual	20007
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.18)	Visual	20008

4.3.2.21 Rocket, practice, 35MM subcaliber, M73 (see 3.6 and dwg. 10242725, prior to assembly of head and closure assembly).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 0.40 Percent		
101.	Propellant broken, cracked, chipped or missing	Visual	21001
102.	Inverted suspension plate	Visual	21002
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.18)	Visual	21003

4.3.2.22 Rocket, practice, 35MM subcaliber: M73 (see dwg. 10242725).

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Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	100 Percent Inspection		
101.	Weight of complete round	Gage	22001
Major:	AQL 0.65 Percent		
102.	Torque of closure to motor incorrect	Gage	22002
103.	Igniter loose	Manual	22003
104.	Safety clip or igniter not sealed properly, bore riding pin or fin broken, or weight adjusting tape improperly positioned	Visual	22004
Minor:	AQL 1.50 Percent		
201.	Marking incorrect or illegible	Visual	22005
202.	Color coded tape improperly positioned	Visual	22006
203.	Evidence of poor workmanship (see 3.18)	Visual	22007

4.3.2.23 Carton, prior to sealing (see dwg. 10242743).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
102.	Orientation of rocket incorrect	Visual	23002
Minor:	AQL 1.50 Percent		
201.	Rocket not packed correctly	Visual	23003
202.	Marking missing, misleading or unidentifiable	Visual	23004
203.	Evidence of poor workmanship (see 3.18)	Visual	23005

4.3.2.24 Carton, sealed (see dwg. 10242743).

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Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 0.40 Percent		
101.	Carton damaged to the extent of exposing components	Visual	24001
102.	Orientation of carton incorrect	Visual	24002
Minor:	AQL 2.50 Percent		
201.	Center band strap on sealing strip not properly attached	Visual	24003
202.	Sealing strips missing	Visual	24004
203.	Carton placed into barrier bag or sealed improperly	Visual	24005
204.	Marking missing, misleading or unidentifiable	Visual	24006
205.	Evidence of poor workmanship (see 3.18)	Visual	24007

4.3.2.25 Box, Packing, prior to closing (see dwg. 10242697).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 0.40 Percent		
101.	Liners not in proper position	Visual	25001
102.	Barrier bags improperly oriented	Visual	25002
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.18)	Visual	25003

4.3.2.26 Box, packing, sealed (see dwg. 10242697).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 0.40 Percent		
101.	Board broken or split	Visual	26001
102.	Binding wire missing or loose	Visual	26002
Minor:	AQL 1.00 Percent		
201.	Marking missing, misleading or unidentifiable	Visual	26003
202.	Evidence of poor workmanship (see 3.18)	Visual	26004

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4.3.3 Testing.

4.3.3.1 First article sample. The first article sample shall be examined 100 percent for all of the characteristics in 4.3.2, and then subjected to the tests listed below, using test equipment in accordance with 4.3.4. Rejection of the first article sample shall be in accordance with 4.2.3.

4.3.3.1.1 Hydrogen embrittlement (see 3.7) safety clip, Code No. 04007; and weight-spring, Code No. 07007. Ten (10) of the safety clips and weight-springs in the first article sample shall be tested as specified in 4.4.1.

4.3.3.1.2 Protective finish (see 3.9) closure, Code No. 03013; safety clip, Code No. 04008; weight-inertia, Code No. 05008; holder primer, Code No. 06008; Weight-spring, Code No. 07008; firing pin, Code No. 08010; plate suspension, Code No. 12009; and motor case, Code No. 13009. Five (5) of each of the above listed components in the first article sample shall be tested as specified in 4.4.2.

4.3.3.1.3 Yield strength (see 3.10.1 and 3.11.1) closure, Code No. 03014; and motor case, Code No. 13010. Ten (10) of the closures and motor cases in the first article sample shall be tested as specified in 4.4.3.

4.3.3.1.4 Ultimate strength (see 3.10.2 and 3.11.2) closure, Code No. 03015; and motor case, Code No. 13011. The ten (10) closures and motor cases used for the yield strength test above should be tested as specified in 4.4.4. Samples used in this test shall be scrapped.

4.3.3.1.5 Resistance of igniter flash tube to stress cracking (see 3.12) Code No. 27001. The fifty (50) flash tubes in the first article sample shall be tested as specified in 4.4.5.

4.3.3.1.6 Resistance to disassembly of primer and holder assembly (see dwg. 10242702) Code No. 09003. Ten (10) of the primer holder assemblies in the first article sample shall be tested as specified 4.4.6.

4.3.3.1.7 Strength of assembly, igniter and primer molded assembly (see 3.13.1) Code No. 20009. Twenty (20) of the igniter and primer molded assemblies in the first article sample shall be tested as specified in 4.4.7.

* 4.3.3.1.8 Air pressure test of lid seal (see dwg 10242696) Code No. 20010. Ten (10) of the igniter and primer molded assemblies from 4.3.3.1.7 shall be tested as specified in 4.4.8.

4.3.3.1.9 Moisture content of black powder (see 3.3) Code No. 20011. Black powder from ten (10) of the igniter and primer molded assemblies from 4.3.3.1.7 shall be tested as specified in 4.4.9.

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4.3.3.1.10 Sensitivity of flash mix (see 3.14) Code No. 02005. The ten (10) loading head assemblies in the first article sample shall be tested as specified in 4.4.10.

4.3.3.1.11 Radiographic examination of closure assembly (see 3.15); safety clip incomplete or improperly assembled Code No. 11006; or assembly latched, Code No. 11007. The ten (10) closure assemblies and all of the complete rounds in the first article sample shall be examined as specified in 4.4.11.

4.3.3.1.12 Contour of motor case nozzle approach angle (see dwg. 10242706) Code No. 13012. The five (5) motor cases in the first article sample, tested in 4.3.3.1.2, shall be tested as specified in 4.4.12.

4.3.3.1.13 Static functioning of igniter (see 3.13.2) peak pressure below minimum, Code No. 20012; peak pressure over maximum, Code No. 20013; rise time less than minimum specified Code No. 20014; rise time exceeds the specified maximum, Code No. 29015, body fails to remain assembled in motor case, Code No. 20016; duds, Code No. 20017. Twenty (20) of the igniter and primer molded assemblies in the first article sample shall be tested as specified in 4.4.13; ten (10) at -10°F and ten (10) at +135°F.

4.3.3.1.14 Waterproofness of igniter (see 3.13.3) duds, Code No. 20018; rise time exceeds maximum, Code No. 20019. Twenty (20) of the igniter and primer molded assemblies in the first article sample shall be tested as specified in 4.4.14.

* 4.3.3.1.15 Flight functioning of complete round (see 3.16) failure to fire on first attempt, Code No. 22008; hangfire in excess of 3 seconds, Code No. 22009; hangfire in excess of 1 second but less than 3 seconds, Code No. 22010; motor case or closure failure, Code No. 22011; premature head function, Code No. 22012; head fails to function on target impact, Code No. 22013; accuracy at +70°F, Code NO. 22014; rocket fails to impact target, code No. 22015; head fails to function on graze impact, Code No. 22016; rocket motor burns past muzzle of launcher, Code No. 22017. The one hundred thirty (130) complete rounds in the first article sample shall be tested as follows:

4.3.3.1.15.1 Accuracy at +70°F. Forty (40) rounds shall be tested as specified in 4.4.15.

4.3.3.1.15.2 Accuracy at -10°F. Twenty (20) rounds shall be tested as specified in 4.4.15.

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4.3.3.1.15.3 Accuracy at +135°F. Twenty (20) rounds shall be tested as specified in 4.4.15.

4.3.3.1.15.4 Functioning at +70°F after water immersion. Fifteen (15) rounds shall be tested as specified in 4.4.16.

4.3.3.1.15.5 Head function on graze impact at +70°F. Fifteen (15) rounds shall be tested as specified in 4.4.17.

4.3.3.1.15.6 Handling safety and propellant contamination (see 3.16.5 and dwg 10242725); head functions from impact of five foot drop, Code No. 22020; primer or igniter functions from impact of five foot drop, Code No. 22021. Twenty (20) rounds shall be tested as specified in 4.4.18. Samples used in this test shall be scrapped.

4.3.3.2 Regular production testing. Sampling for the following listed characteristics shall be applied to "inspection lots" (see 4.3.1) and as such applies to heat treatment batches, heats of metal, etc. Test equipment used in performance of the applicable tests shall be in accordance with 4.3.4.

4.3.3.2.1 Hydrogen embrittlement (see 3.7) safety clip, major defect, Code No. 04007; and weight-spring, major defect, Code No. 07007. A sample of ten (10) safety clips and springs shall be selected from each heat treatment batch and tested as specified in 4.4.1. The heat treatment batch represented by one or more defective components shall be rejected.

4.3.3.2.2 Protective finish (see 3.9) major defects; closure, Code No. 03013; safety clip, Code No. 04008; weight-inertia, Code No. 05008; holder primer, Code No. 06008; weight-spring, Code No. 07008; firing pin, Code No. 08010; plate suspension, Code No. 12009; and motor case, Code No. 13009. If visual inspection of above parts reveals discrepancies in protective finish, suspected lots of these parts shall be submitted to first article sample test as indicated in para 4.3.3.1.2. Deficient lots of any component shall be rejected and each subsequent lot of that component shall be tested until an acceptable lot is produced.

4.3.3.2.3 Yield strength of closure and motor case (see 3.10.1 and 3.11.1) major defects; closure Code No. 03014; and motor case, Code No. 13010. This sampling shall be in accordance with MIL-STD-105 Table II-A, AQL 0.065. Rejected lots shall revert to 100% inspection. This test shall be performed as specified in 4.4.3.

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* 4.3.3.2.4 (Deleted in this revision.)

4.3.3.2.5 Resistance of igniter flash tube to stress cracking (see 3.12) major defect; Code No. 27001. A sample of fifty (50) flash tubes shall be selected from each lot and subjected to the test specified in 4.4.5. The lot shall be rejected if one or more flash tubes is classed defective.

4.3.3.2.6 Resistance to disassembly of primer and holder assembly (see dwg. 10242702) major defect; Code No. 09003. This test shall be performed one hundred percent. This test shall be performed as specified in 4.4.6. Defective assemblies shall be rejected and removed from the lot.

4.3.3.2.7 Strength of assembly, igniter and primer molded assembly (see 3.13.1) major defect; Code No. 20009. A sample of fifty (50) igniter and primer molded assemblies shall be selected from each lot and subjected to the test specified in 4.4.7. The lot shall be rejected if two or more assemblies are classed defective. If one assembly is classed defective, a second sample of one hundred (100) assemblies shall be selected from the lot and subjected to the test. The igniter and primer molded assembly lot shall be rejected upon encountering one defective assembly and at the option of the contractor testing may be stopped.

4.3.3.2.8 Strength of lid seal (see dwg 10242744) major defect, Code No. 200010. Fifty igniter loading subassemblies shall be tested as specified in 4.4.8. The igniter loading subassembly lot shall be rejected if one or more assemblies are classed defective.

4.3.3.2.9 Moisture content of the black powder (see 3.3) Major Defect Code No. 27001. A sample of approximately three grams of black powder shall be selected from each container when loading from the container is initiated, and every four hours thereafter, until the contents of the container are depleted. The sample shall be divided into three approximately equal specimens and each specimen tested as specified in 4.4.9. If one or more specimens fail to conform to the requirement specified, the quantity loaded from that container since the last test shall be rejected.

* 4.3.3.2.10 Sensitivity of flash mix (see 3.14) Code No. 02005. A blended flash mix specimen shall be selected from four (4) heads during each two hours production and subjected to the test specified in 4.4.10. Each two hours production shall be rejected if one or more specimen is classed defective.

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4.3.3.2.11 Radiographic examination of closure assembly (see 3.15) major defects; safety clip incomplete or improperly assembled, Code No. 11006; or assembly latched, Code No. 11007. This examination shall be performed one hundred percent. This test shall be performed as specified in 4.4.11. Assemblies classed defective shall be rejected and removed from the lot.

4.3.3.2.12 Contour of motor case nozzle approach angle (See dwg 10242706) major defect; Code No. 13012. A sample of five (5) motor cases per MIL-STD-105 Level II shall be selected from each motor case lot and examined as specified in 4.4.12. The motor case lot shall be rejected if one or more specimens is classed defective. In case of a rejected lot, a second sample of 16 motor cases shall be selected from the motor case lot and examined as specified in 4.4.12. The motor case lot shall be rejected if two samples are rejected from the 16 samples.

4.3.3.2.13 Static functioning of igniter (see 3.13.2) major defects; peak pressure below minimum, Code No. 20012; peak pressure over maximum, Code No. 20013, rise time less than minimum specified, Code No. 20014; rise time exceeds the specified maximum, Code No. 20015; body fails to remain assembled in motor case, Code No. 20016; and duds, Code No. 20017. A sample of eighty (80) igniter and primer molded assemblies shall be selected from each lot and subjected to the test specified in 4.4.13; forty (40) at -10°F and forty (40) at +135°F. The igniter and primer molded assembly lot shall be rejected when:

two or more duds are observed, or if
one or more igniter cups are ejected, or ruptured, or
two or more igniters exhibit nonconforming rise time, or if
three or more igniters exhibit nonconforming peak pressure, or
one or more igniters exhibit pressure below 250 psi.

4.3.3.2.14 Waterproofness of igniter (see 3.13.3) major defects; duds, Code No. 20018; rise time exceeds maximum, Code No. 20019. A sample of forty (40) igniter and primer molded assemblies shall be selected from each lot and subjected to the test specified in 4.4.14. The igniter and primer molded assembly lot shall be rejected when:

two or more duds are observed, or
two or more assemblies exhibit excessive rise time.

If one of either of the above defects are observed, a second sample of 40 igniter and primer molded assemblies shall be selected and tested. Testing of the second sample shall be stopped and the lot rejected if two or more duds are observed, or two or more assemblies exhibit excessive rise time in the total sample of 80 items.

* 4.3.3.2.15 Flight functioning of complete round (see 3.16.1) failure to fire on first attempt, Code No. 22008; hangfire in excess of 3 seconds, Code No. 22009; hangfire in excess of 1 second but less than 3 seconds, Code No. 22010; motor case or closure failure, Code No. 22011; premature head function, Code No. 22012; head fails to function on target impact, Code No. 22013; accuracy at +70°F, Code No. 22014; rocket fails to impact target, Code No. 22015; head fails to function on graze impact, Code No. 22016; rocket motor burns past muzzle of launcher, Code No. 22017 (for information only). A sample of ninety five (95) complete rockets shall be selected from each lot and tested as indicated in the following sections.

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4.3.3.2.15.1 Accuracy at +70°F. Forty (40) rounds shall be tested as specified in 4.4.15.

4.3.3.2.15.2 Accuracy at -10°F. Twenty (20) rounds shall be tested specified in 4.4.15.

4.3.3.2.15.3 Accuracy at +135°F. Twenty (20) rounds shall be tested as specified in 4.4.15.

4.3.3.2.15.4 Functioning at +70°F after water immersion. Fifteen (15) rounds shall be tested as specified in 4.4.16.

* 4.3.3.2.16 Reject criteria. The complete round lot shall be rejected if:

three rockets fail to launch on the first attempt, or if
one rocket hangfires in excess of 3 seconds, or if
two rockets hangfire in excess of 1 second but less than 3 seconds, or if
two closures or motor cases fail, or if
one head functions prematurely, or if
three heads fail to function on target impact, or if
in the combined sample of 55 rockets tested in 4.3.3.2.15.1 and 4.3.3.2.15.4 six rockets miss the 10 foot by 10 foot target, or when considering the coordinates of impact of each rocket that impacts the target during the +70°F accuracy phase, the average center of impact plus or minus the product of 1.5 and the sample standard deviation exceed the limits of the 10 ft by 10 ft target. The sample standard deviation shall be calculated as specified in 6.4.

4.3.4 Inspection equipment. The inspection equipment required to perform the examinations and tests required by this section is identified by Equipment Tabulation ET-10242725.

4.4 Test methods and procedures. The test methods and procedures described in each of the following tests represent only one method and one procedure which has been proved in use and is approved by the Government for Government acceptance purposes. Producers and manufacturers have the option of using substitute methods and procedures only after obtaining Government approval. Government approval will be based on an evaluation to determine that the substitute method and procedure will result in an equal or better quality evaluation than that specified herein.

4.4.1 Embrittlement of safety clip and inertia spring. The inertia spring shall be extended along its longitudinal axis until it no longer resumes its pre-stressed length. The pointed end of the safety clip shall be deformed in a direction to line up with the adjacent straight section until it no longer returns to its pre-stressed configuration. The component shall be classed defective if it breaks during this test.

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4.4.2 Protective finish. The components shall be tested as specified in ASTM-B117 and the requirements specified herein. The component shall be classed defective if exposure to the salt spray for the specified time produces white corrosion products of cadmium.

4.4.3 Yield strength of motor case and closure. The motor case shall be assembled to a test fixture by means of the threads only. The fixture threads shall represent the minimum metal condition of the closure and shall seal the nozzle throat as close to the throat as possible and still retain the seal. The closure shall be assembled to a fixture which simulates the motor case/closure interface by means of the threads only. After the specified yield strength pressure has been applied, the component shall be examined for cracks, splits, ruptures, or stripped threads. The dimensional examination required by 4.3.2.3 and 4.3.2.13. shall be performed on the lot subsequent to this test.

4.4.4 Ultimate strength of motor case and closure. The component shall be assembled to the applicable test fixtures similar to those in 4.4.3 and an internal hydrostatic pressure of increasing magnitude shall be applied until failure occurs, or 25,000 minimum psi is achieved. The pressure at which failure occurred shall be recorded. Samples used in this test shall be scrapped.

4.4.5 Resistance of igniter flash tube to stress cracking. The flash tube shall be heated in a container or in water heated to the required temperature for approximately ten (10) minutes. The tube shall be removed from the heated environment, bent 180 degrees and placed into the stress fixture which will hold the open ends parallel and at the specified separation. The flash tube and fixture shall then be submerged in the specified stress cracking reagent at the required temperature. After being submerged for the specified time, the flash tube shall be examined for cracks, splits and breaks.

4.4.6 Resistance to disassembly, primer from primer holder. The minimum specified load shall be applied to the primer through the flash hole. Observation of inadvertant primer functioning or primer disassembly shall class the assembly defective.

4.4.7 Strength of assembly. The required force shall be applied to the primer block on one end and to the motor case at the other end along the longitudinal axis of the flash tube in a manner to pull the assembly apart. The assembly shall be classed defective if either end of the flash tube becomes disassembled.

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* 4.4.8 Air pressure test of lid seal. The flash tube of the igniter loading subassembly shall be attached to a source of air pressure which will apply the required internal pressure to the lid and lid seal. The assembly will then be submerged under two to three inches of water for at least 10 seconds while retaining the prescribed air pressure. The lid seal shall be classed defective if a leak is evidenced by bubbles emerging from around the lid.

4.4.9 Moisture content of black powder. The black powder specimen shall be placed in a tared weighing dish and covered with a tared cover. The weight shall be recorded to the nearest milligram, the weight of the dish and cover subtracted and the result recorded as W. The cover and the uncovered dish containing the specimen shall be placed in an oven maintained at 70 plus 5 degrees Centigrade for five (5) hours ± 0.5 hour. The cover shall then be placed over the dish and specimen, transferred to a dessiccator, cooled to room temperature, and the weight recorded to the nearest milligram. The weight of the dish and cover shall be subtracted from this weight and the result recorded as D. The percentage of moisture is calculated as:

$$\text{Percentage moisture} = \frac{W - D}{W} (100)$$

4.4.10 Sensitivity of flash mix. The die cap of the specified impact test apparatus shall be filled level with the blended flash mixture, the cap assembled, the assembly placed in the anvil, the vented plug centered on the cap and the weight dropped from the required height to impact the plug. The specimen shall be classed defective if it detonates or burns.

4.4.11 Radiographic examination. The closure assembly with the safety clip assembled and sealed shall be radiographically examined by positioning the assembly so that the longitudinal axis of the safety clip hole is parallel to the film plane. The assembly shall be classed defective if it fails to meet any of the specified requirements.

4.4.12 Contour of motor case nozzle approach angle. The motor case shall be sectioned along the plane of the longitudinal axis and the sectioned half examined with the aid of a comparator and template depicting the minimum and maximum throat and approach angle contour. Non conforming motor cases shall be classed defective.

4.4.13 Static functioning of igniter and primer molded assembly. The igniter molded cup shall be sealed in the motor case nozzle as shown on dwg 10242725 and conditioned at the required temperature ± 5°F for a minimum of six hours. The assembly shall be assembled to simulated closure containing an insert representing the volume of the propellant charge assembly and suspension plate. The primer shall be impacted by an appropriate firing pin. The point should not puncture the primer to allow venting of the igniter. Two simultaneous time traces shall be produced by the equipment and the average of both traces shall be utilized in the

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reduction of data. In the event one channel fails, the remaining result shall be considered as absolute and adequate. The assembly shall be functioned in not less than two (2) minutes and in not more than three (3) minutes after removal from the conditioning chamber. The results shall be recorded, and the traces compared to the requirements. Photographs of the pressure/time of each trace in the first twenty (20) lots shall be recorded. A typical sample of 3 traces shall be photographed from each subsequent tenth lot at each test temperature for information purposes and the test data shall be provided to the Government in accordance with the applicable contract.

4.4.14 Waterproofness of Igniter. The igniter and primer molded assembly shall be prepared for test by seating and sealing the igniter in the motor case, as shown on dwg 10242725, and waterproof sealing the threaded end of the motor case. The assembly shall be removed from the water and excess water removed by wiping only. The assembly shall be assembled to the test equipment utilized in 4.4.13 and functioned. The required results shall be recorded from the trace, and compared to the requirements.

* 4.4.15 Flight Functioning. The target for the 64 degree functioning phase shall be 3/16 inch minimum thickness of steel. The target surface shall form an angle of sixty-four (64) degrees with the vertical, and shall be located forty-five meters from the launcher. For the accuracy phase, a ten (10) foot square target shall be centered on a vertical target no smaller than twelve (12) feet square and placed two hundred (200) meters, ± one (1) meter from the muzzle of the launcher. The centered square shall be marked off in a grid consisting of horizontal (x axis) and vertical (y axis) lines from zero (0) to one hundred twenty (120) inches. Zero shall be the extreme left edge and lower edge of the ten (10) foot square. Points of impact shall be measured to the nearest one half (1/2) inch. The launcher shall be bore sighted at the center of the target (x=60, y=60) and raised fifty-four (54) mils for launching at +70°F; 55.7 mils for launching at -10°F; and 52.8 mils for launching at +135°F. The complete rounds shall be conditioned at the required temperature, ±2°F, for a minimum of twelve (12) hours and then fired from the rigid steel tube test launcher. Launch shall be accomplished in not less than one (1) minute and not more than three (3) minutes after removal from the conditioning chamber. If launch will not be effected within the three (3) minute limit, the round shall not be fired, but shall be returned for one hour of conditioning for each two (2) minutes in excess of three (3) minutes out of the conditioning chamber. In addition to observing and recording the test results for the specified requirements, the velocity of the round shall be recorded at between five (5) and fifteen (15) feet from the launcher muzzle. Flight tests shall not be conducted in winds exceeding 15 knots (17.25 miles per hour). Rockets to be fired for accuracy at 70°F shall be visually inspected prior to firing. Any rocket with broken or missing parts shall not be fired but shall be replaced by selecting suitable rounds

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from the 135°F tests. These rounds require temperature conditioning in accordance with applicable procedures for the 70°F accuracy phase.

4.4.16 Flight after water immersion. The complete round shall be submerged in water to the required depth with the water maintained within the specified temperature limits. After the required time has elapsed, the rocket shall be removed from the water, placed in the test launcher and fired at the target as specified in 4.4.15. The test launcher shall be elevated to the 54 mil setting.

4.4.17 Head functioning at 64 degrees. The rocket shall be conditioned at 70° + 20°F and fired from the test launch tube at the target as specified in 4.4.15 and positioned at 64 degrees from the vertical. The target material shall be the same as that used in the accuracy phase and placed 45 meters from the muzzle end of the launch tube. The velocity of the round shall be recorded. Observations shall be made of rocket performance for comparison against each of the requirements.

4.4.18 Five-foot drop bare. The rocket, with the safety clip removed, shall be dropped once on the tail, then once on the head. The rockets used in this test shall be destroyed. The rocket shall be classed defective if, during drop, it becomes propulsive, or if the head detonates.

5. PACKAGING

5.1 Preservation, packaging, packing and marking. Preservation, packaging, packing and marking will be in accordance with drawings 10242743 and 10242697.

6. NOTES

6.1 Ordering data. Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards. Data cards shall be prepared for each lot in accordance with the information specified in MIL-STD-1167.

c. Provisions for submission of first article samples.

d. Copies of pressure/time trace photos of igniter static functioning test and rocket motor and closure ultimate strength test shall be forwarded to the Government in accordance with the appropriate contract.

6.2 Inspection Code numbers. The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are used to correlate the characteristics cited on the Equipment Lists with the inspections listed in this specification. They should also be cited as references on drawings of equipment designs submitted by the contractor (see 6.3.3).

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6.3 Inspection equipment designs. Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers (under the "Method of Inspection" headings in Section 4). Design responsibility for all other inspection equipment is assigned to the contractor. Equipment to be designed by the contractor is designated "CD". The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3, should he elect to do so.

6.3.1 SIE designs. SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or specification control drawing, as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor designs. Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public.) Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government, and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.3.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements List DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five-digit defect code number (or other specific identifying information) from Section 4 of this specification shall be referenced on each contractor design, together with the component or assembly drawing number and revision letter to which the specific design applies.

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6.3.3 Submission of Designs for Approval. Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs.

6.4 Calculation of sample standard deviation. The sample standard deviation will be calculated using the following formula or its equivalent:

$$S = \sqrt{\frac{\Sigma (X - \bar{X})^2}{n - 1}}$$

$\bar{X}$ = Average value for the sample
X = Numerical value for individual observations
s = Sample standard deviation
n = Sample size

6.5 Submission of first article sample. Instructions as to the location for the evaluation of first article sample shall be obtained from the contracting officer. Delivery of test data, hardware and inspection records shall be in accordance with appropriate contract documents.

6.6 Testing of Government-owned initiators. The Contracting Officer will arrange to have the igniter contractor reimbursed for expenses incurred in the performance of the check test for deterioration of primers when the contracator is directed to perform the test.

6.7 Changes to previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modification, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Army - MI

Review Activity:
Army - PA

Preparing Activity:
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OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-R-50858C(MI) ROCKET, PRACTICE, 35MM, SUBCALIBER, M73, PARTS FOR LOADING AND ASSEMBLING

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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